

Practice Question Set

1. What are comments in C?
 2. What are the two types of comments in C?
 3. Why are comments used?
 4. What are control instructions?
 5. What are the four types of control instructions in C?
 6. What is the Golden Rule of truth in C?
 7. What is an `if` statement?
 8. What is an `if-else` statement?
 9. What is the Single-Line Rule in `if-else` statements?
 10. What are loops? Name the three types of loops.
 11. What is a while loop?
 12. What is a do-while loop?
 13. What is a for loop?
 14. What is the purpose of the `break` statement?
 15. What is a switch-case statement?
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Important Interview/Viva Questions

16. Difference between single-line and multi-line comments.
17. Difference between `if` and `if-else`.

18. Difference between while, do-while, and for loops.
 19. Why is the do-while loop called an exit-controlled loop?
 20. What is an infinite loop?
 21. Why is a semicolon mandatory after `while ()` in a do-while loop?
 22. Difference between break and continue.
 23. Difference between if-else and switch-case.
 24. What are the rules of switch-case?
 25. Why is break used in switch-case statements?
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Answers :

1. What are comments in C?

Answer:

Comments are non-executable statements used to provide notes and explanations in programs.

The compiler ignores comments completely.

2. What are the two types of comments?

Answer:

Single-Line Comment

```
// This is a comment
```

Multi-Line Comment

```
/* This  
   is a  
   comment */
```

3. Why are comments used?

Answer:

Comments are used for:

- Documentation
- Explanation of code
- Improving readability
- Making maintenance easier

4. What are control instructions?

Answer:

Control instructions control the flow of execution of a program.

5. What are the four types of control instructions?

Answer:

1. Decision Control Instructions
2. Iterative Control Instructions
3. Switch-Case Control Instructions
4. Goto Control Instructions

6. What is the Golden Rule of truth in C?

Answer:

- Zero (0) → False
- Any non-zero value → True

Examples:

0 → False
1 → True

-5 → True
100 → True

7. What is an if statement?

Answer:

The `if` statement executes a block only when the condition is true.

Syntax:

```
if (condition)
{
    // statements
}
```

8. What is an if-else statement?

Answer:

It provides two execution paths.

Syntax:

```
if (condition)
{
    // True block
}
else
{
    // False block
}
```

9. What is the Single-Line Rule?

Answer:

If only one statement is present inside an `if` or `else`, braces are optional.

Example:

```
if (a > 0)
    printf("Positive");
```

However, using braces is recommended.

10. What are loops?

Answer:

Loops repeatedly execute a block of code while a condition remains true.

Types:

- while loop
 - do-while loop
 - for loop
-

11. What is a while loop?

Answer:

A while loop is an entry-controlled loop.

The condition is checked before executing the loop body.

Syntax:

```
while (condition)
{
    // statements
}
```

12. What is a do-while loop?

Answer:

A do-while loop is an exit-controlled loop.

The loop body executes at least once.

Syntax:

```
do
{
```

```
    // statements
}
while(condition);
```

13. What is a for loop?

Answer:

A for loop combines initialization, condition, and update in one line.

Syntax:

```
for(initialization;condition;update)
{
    // statements
}
```

14. What is the purpose of break?

Answer:

`break` immediately terminates a loop or switch statement.

Example:

```
if (age==18)
{
    break;
}
```

15. What is a switch-case statement?

Answer:

A switch statement selects one block among multiple choices.

Syntax:

```
switch(expression)
{
case 1:
    statements;
    break;
```

```

case 2:
    statements;
    break;

default:
    statements;
}

```

Important Interview Answers

16. Difference between Single-Line and Multi-Line Comments

Single-Line	Multi-Line
Uses //	Uses /* */
One line	Multiple lines
Short notes	Long descriptions

17. Difference between if and if-else

if	if-else
One block	Two blocks
Executes only if condition is true	Executes one block when true and another when false

18. Difference between while, do-while, and for loops

while	do-while	for
Entry-controlled	Exit-controlled	Entry-controlled
Condition checked first	Condition checked last	Initialization, condition, update together
May execute zero times	Executes at least once	Counter-driven loop

19. Why is do-while called an exit-controlled loop?

Answer:

Because the condition is checked after executing the loop body.

Hence the body executes at least once.

20. What is an infinite loop?

Answer:

A loop that never terminates is called an infinite loop.

Example:

```
while(1)
{
    printf("Hello");
}
```

21. Why is semicolon mandatory after while() in a do-while loop?

Answer:

Because the syntax of the do-while loop requires it.

```
do
{
}
while(condition);
      ↑
    Mandatory
```

22. Difference between break and continue

break	continue
Terminates loop	Skips current iteration
Exits loop completely	Goes to next iteration

23. Difference between if-else and switch-case

if-else	switch-case
Uses conditions	Uses constant values
Supports relational operators	Only integer or character expressions
Better for complex conditions	Better for multiple choices

24. What are the rules of switch-case?

Answer:

1. Expression must be integer or character type.
2. Case labels must be constants.
3. Duplicate cases are not allowed.
4. Variables cannot be used as case labels.
5. Default block is optional.

25. Why is break used in switch-case?

Answer:

`break` prevents fall-through and exits the switch statement.

Without `break`, execution continues into the next case.